Manage sprints and capacity

Teams build the sprint backlog during the sprint planning meeting, typically held on the first day of the sprint. Each sprint corresponds to a time-boxed interval that supports the team's ability to work using Agile processes and tools. During the planning meeting, the product owner works with the team to identify those stories or backlog items to complete in the sprint.

Planning meetings typically consists of two parts. In the first part, the team and product owner identify the backlog items that the team feels it can commit to completing in the sprint, based on experience with previous sprints. These items get added to the sprint backlog. In the second part, the team determines how it will develop and test each item. They then define and estimate the tasks required to complete each item. Finally, the team commits to implementing some or all the items based on these estimates.

The sprint backlog should contain all the information the team needs to successfully plan and complete work within the time allotted without having to rush at the end. Before planning the sprint, you'd want to have created, prioritized, and estimated the backlog and defined the sprints.

1. In the vertical navigational pane of the Azure DevOps portal, select the **Boards** icon and, in the list of the **Boards** items, select **Sprints**.
2. On the **Taskboard** tab of the **Sprints** view, in the toolbar, on the right hand side, select the **View options** symbol (directly to the left of the funnel icon) and, in the **View options** dropdown list, select the **Work details** entry. Select **Sprint 2** as filter.

Note: The current sprint has a pretty limited scope. There are two tasks in the **To do** stage. At this point, neither task has been assigned. Both show a numeric value to the right of **Unassigned** entry representing the remaining work estimate.

3. Within the **ToDo** Column, notice the Task Item **Add page for most recent tutorials**, select the **Unassigned** entry and, in the list of user accounts, select your account to assign the task to yourself.
4. Select the **Capacity** tab of the **Sprints** view.

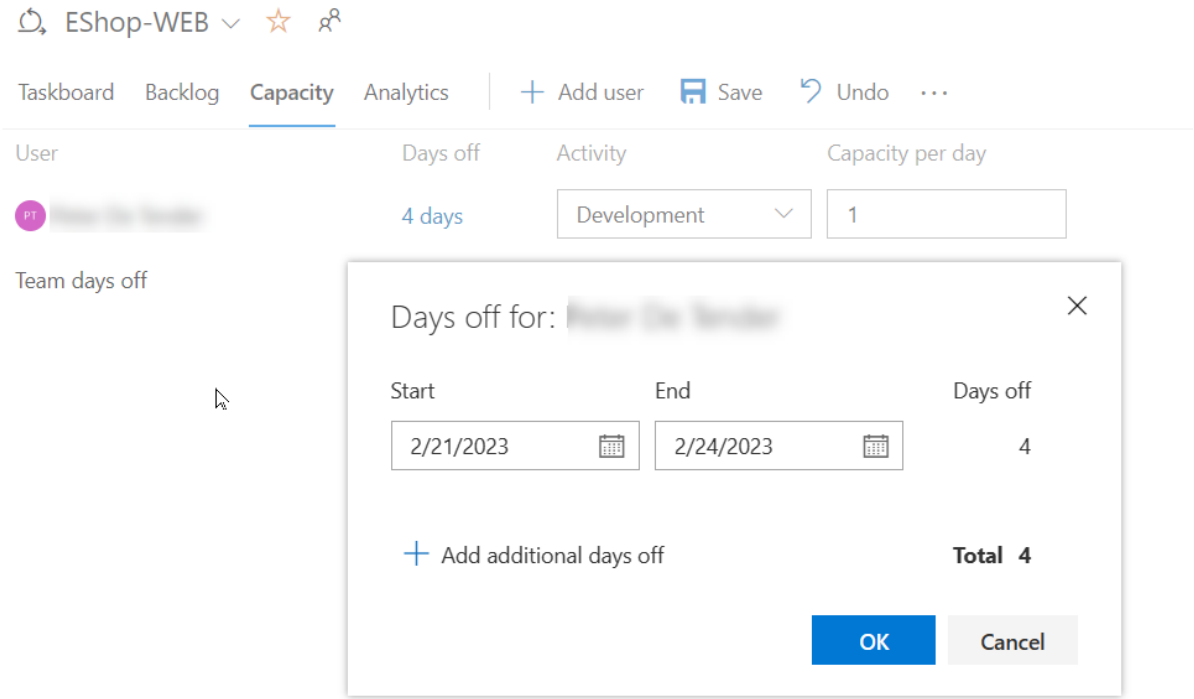
Note: This view enables you to define what activities a user can take on and at what level of capacity.

5. On the **Capacity** tab of the **Sprints** view, for your user account, set the **Activity** field to **Development** and, in the **Capacity per day** textbox, type **1**. Then select **Save**.

Note: Let's assume you're also going to take some vacation. Which should be added to the capacity view too.

6. On the **Capacity** tab of the **Sprints** view, directly next to the entry representing your user account, in the **Days off** column, select the **0 days** entry. This will display a panel where you can set your days off.

7. In the displayed panel, use the calendar view to set your vacation to span four work days during the current sprint (within the next three weeks) and, once completed, select **OK**.



8. Back on the **Capacity** tab of the **Sprints** view, select **Save**.

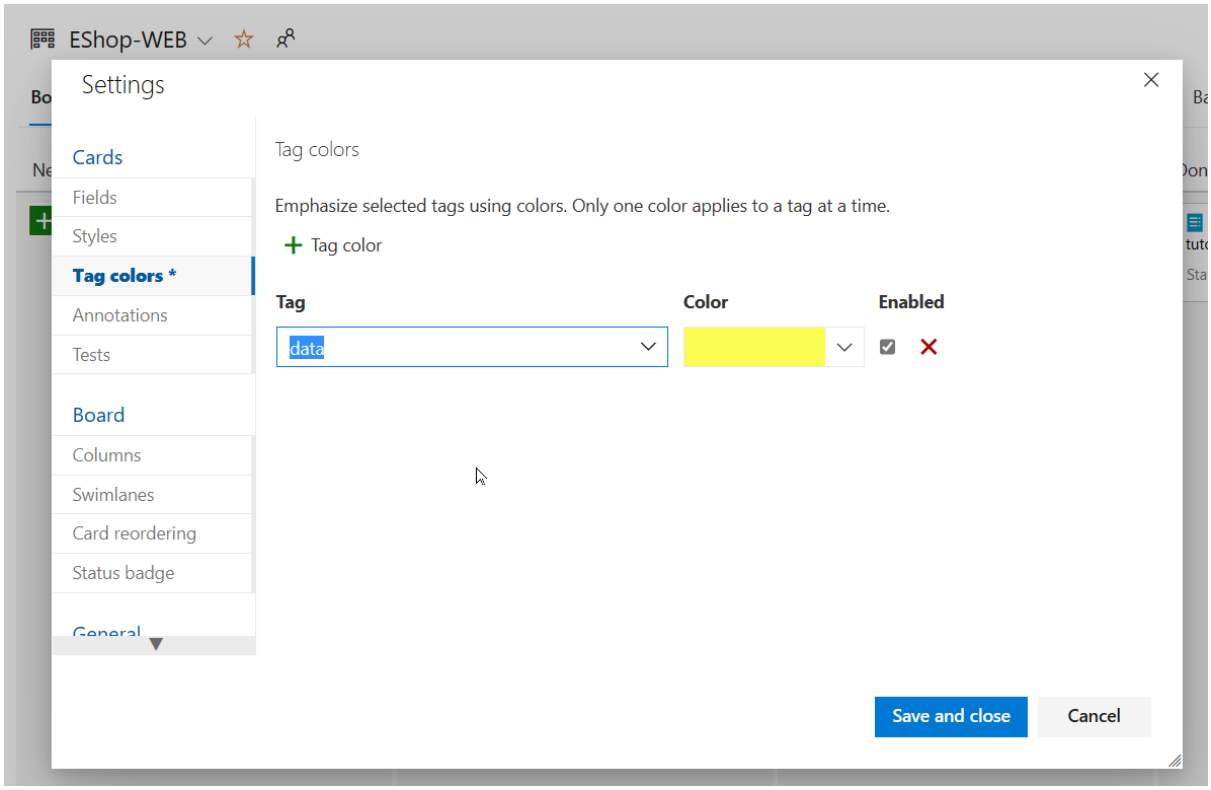
Customize Kanban boards

To maximize a team's ability to consistently deliver high quality software, Kanban emphasizes two main practices. The first, visualizing the flow of work, requires that you map your team's workflow stages and configure a Kanban board to match. The second, constraining the amount of work in progress, requires that you set work-in-progress (WIP) limits. You're then ready to track progress on your Kanban board and monitor key metrics to reduce lead or cycle time. Your Kanban board turns your backlog into an interactive signboard, providing a visual flow of work. As work progresses from idea to completion, you update the items on the board. Each column represents a work stage, and each card represents a user story (blue cards) or a bug (red cards) at that stage of work. However, every team develops its own process over time, so the ability to customize the Kanban board to match the way your team works is critical for the successful delivery.

1. In the vertical navigational pane of the Azure DevOps portal, in the list of the **Boards** items, select **Boards**.
2. On the **Boards** panel, select the **Configure board settings** cogwheel icon (directly to the right of the funnel icon).

Note: The team is emphasizing work done with data, so there is special attention paid to any task associated with accessing or storing data.

3. On the **Settings** panel, select the **Tag colors** tab, select **+ Add tag color**, in the **Tag** textbox, type **data** and select the yellow rectangle.



Note: Whenever a backlog item or bug is tagged with **data**, that tag will be highlighted.

4. Select the **Annotations** tab.

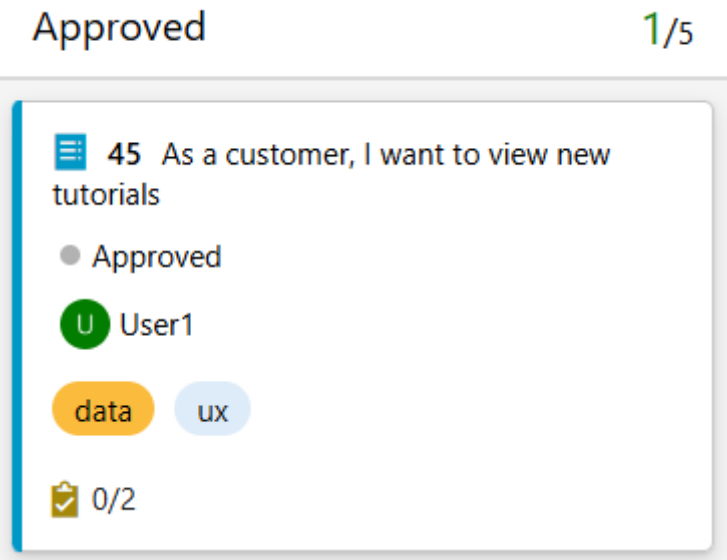
Note: You can specify which **Annotations** you would like included on cards to make them easier to read and navigate. When an annotation is enabled, the child work items of that type are easily accessible by selecting the visualization on each card.

5. Select the **Tests** tab.

Note: The **Tests** tab enables you to configure how tests appear and behave on the cards.

6. Select **Save** to save and close the settings.
7. From the **Board** tab of the **EShop-Web** panel, open the Work Item representing the **As a customer, I want to view new tutorials** backlog item.

8. From the detailed item view, at the top of the panel, to the right of the **0 comments** entry, select **Add tag**.
9. In the resulting textbox, type **data** and press the **Enter** key.
10. Repeat the previous step to add the **ux** tag.
11. Save these edits by selecting **Save and Close**.



Note: The two tags are now visible on the card, with the **data** tag highlighted in yellow as configured.

12. On the **Boards** panel, select the **Configure board settings** cogwheel icon (directly to the right of the funnel icon).
13. On the **Settings** panel, select the **Columns** tab.

Note: This section allows you to add new stages to the workflow.

14. Select **+ Add column**, under the **Column name** label, in the **Name** textbox, type **QA Approved** and, in the **WIP limit** textbox, type **1**

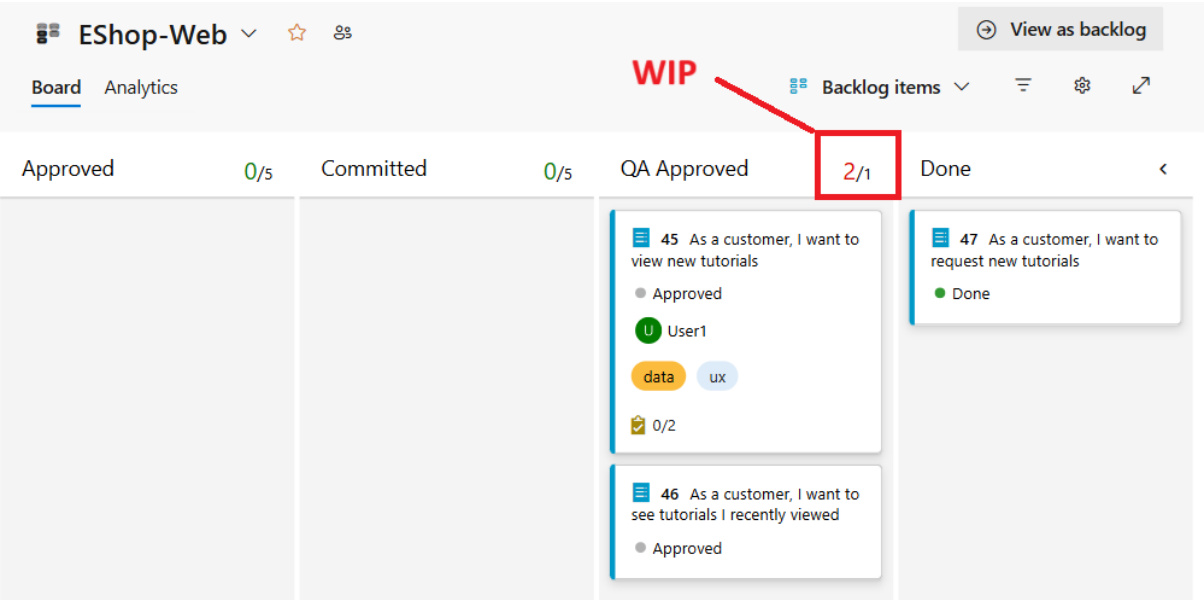
Note: The Work in progress limit of 1 indicates that only one work item should be in this stage at a time. You would ordinarily set this higher, but there are only two work items to demonstrate the feature.

15. Notice the ellipsis next to the **QA Approved** column you created. Select **Move right** twice, so that the QA Approved column gets positioned in-between **Committed** and **Done**.
16. Select **Save** to save and close the settings.

17. On the **Boards portal**, the **QA Approved** column is now visible in the Kanban board view.

18. Drag the **As a customer, I want to see tutorials I recently viewed** work item from the **Committed** stage into the **QA Approved** stage.

19. Drag the **As a customer, I want to view new tutorials** work item from the **Approved** stage into the **QA Approved** stage.



Note: The stage now exceeds its **WIP** limit and is colored red as a warning.

20. Move the **As a customer, I want to see tutorials I recently viewed** backlog item back to **Committed**.

21. On the **Boards** panel, select the **Configure board settings** cogwheel icon (directly to the right of the funnel icon).

22. On the **Settings** panel, return to the **Columns** tab and select the **QA Approved** tab.

Note: A lag often exists between when work gets moved into a column and when work starts. To counter that lag and reveal the actual state of work in progress, you can turn on split columns. When split, each column contains two sub-columns: **Doing** and **Done**. Split columns let your team implement a pull model. Without split columns, teams push work forward, to signal that they've completed their stage of work. However, pushing it to the next stage doesn't necessarily mean that a team member immediately starts work on that item.

23. On the **QA Approved** tab, enable the **Split column into doing and done** checkbox to create two separate columns.

Note: As your team updates the status of work as it progresses from one stage to the next, it helps that they agree on what **done** means. By specifying the **Definition of done** criteria for

each Kanban column, you help share the essential tasks to complete before moving an item into a downstream stage.

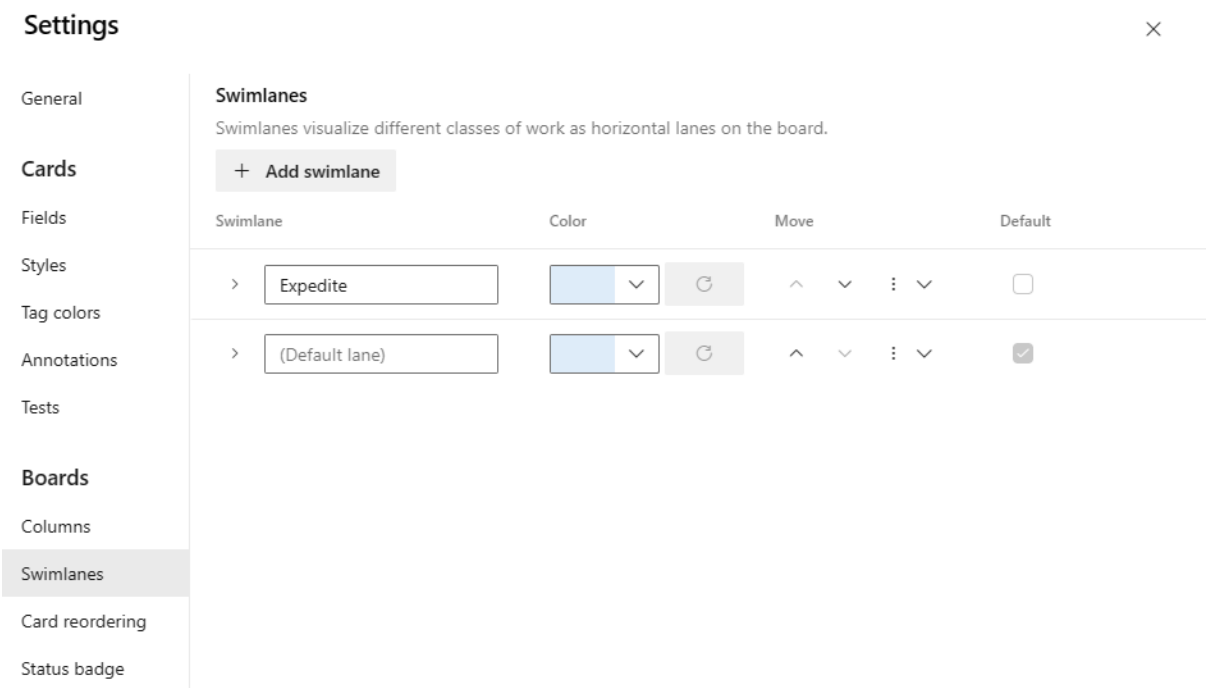
24. On the **QA Approved** tab, at the bottom of the panel, in the **Definition of done** textbox, type **Passes **all** tests**.
25. Select **Save** to save and close the settings.

Note: The **QA Approved** stage now has **Doing** and **Done** columns. You can also select the informational symbol (with letter **i** in a circle) next to the column header to read the **Definition of done**. You may need to refresh the browser to see changes.

26. On the **Boards** panel, select the **Configure boards settings** cogwheel icon (directly to the right of the funnel icon).

Note: Your Kanban board supports your ability to visualize the flow of work as it moves from new to done. When you add **swimlanes**, you can also visualize the status of work that supports different service-level classes. You can create a swimlane to represent any other dimension that supports your tracking needs.

27. On the **Settings** panel, select the **Swimlanes** tab.
28. On the **Swimlanes** tab, select **+ Add swimlane**, directly under the **Swimlane name** label, in the **Name** textbox, type **Expedite**.
29. In the **Color** dropdown list, select the **Green** rectangle.
30. Select **Save** to save and close the settings.



31. Back on the **Board** tab of the **Boards** panel, drag and drop the **Committed** work item onto the **QA Approved | Doing** stage of the **Expedite** swimlane so that it gets recognized as having priority when QA bandwidth becomes available.

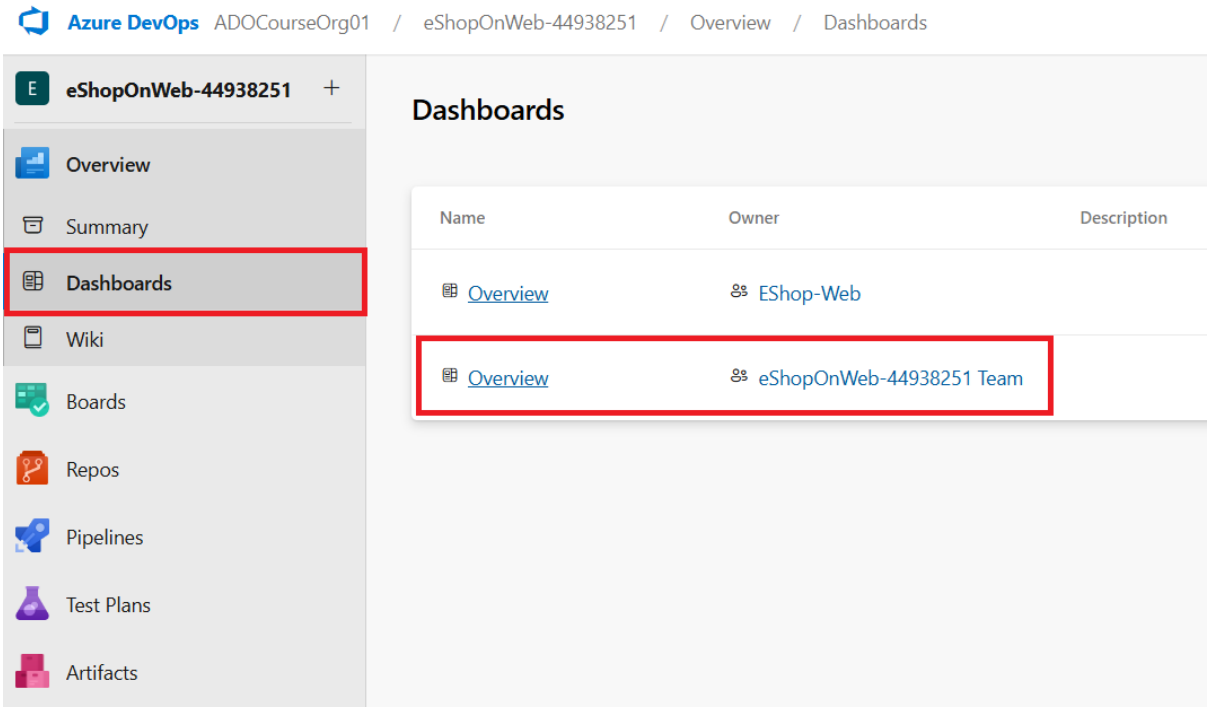
Note: You may need to refresh the browser to make the swimlane visible.

Define dashboards

Dashboards allow teams to visualize status and monitor progress across the project. At a glance, you can make informed decisions without having to drill down into other parts of your team project site. The Overview page provides access to a default team dashboard which you can customize by adding, removing, or rearranging the tiles. Each tile corresponds to a widget that provides access to one or more features or functions.

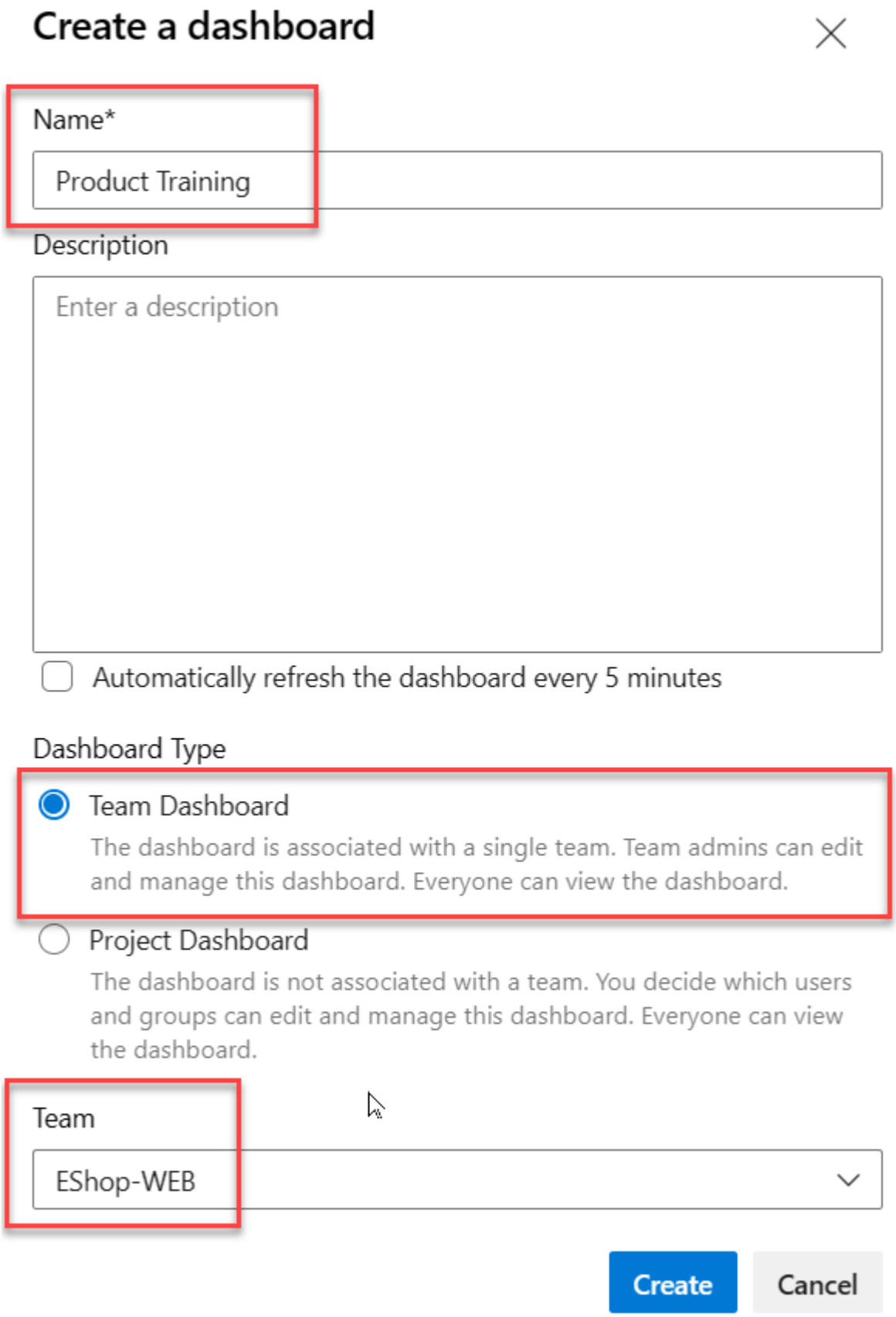
1. In the vertical navigational pane of the Azure DevOps portal, select the **Overview** icon and, in the list of the **Overview** items, select **Dashboards**.

2. Select the **Overview** for the **eShopOnWeb Team** and review the existing dashboard.



3. On the **Dashboards** pane, in the upper-left corner, expand the drop-down menu under the Dashboard's name and select **+ New Dashboard**.

4. On the **Create a dashboard** pane, in the **Name** textbox, type **Product training**, in the **Team** dropdown list, select the **EShop-Web** team, and select **Create**.



5. On the new dashboard pane, select **Add a widget**.

6. On the **Add Widget** panel, in the **Search widgets** textbox, type **sprint** to find existing widgets that focus on sprints. In the list of results, select **Sprint Overview** and select **Add**.

7. In the rectangle representing the newly added widget, select the **Settings** cogwheel icon and review the **Configuration** pane.

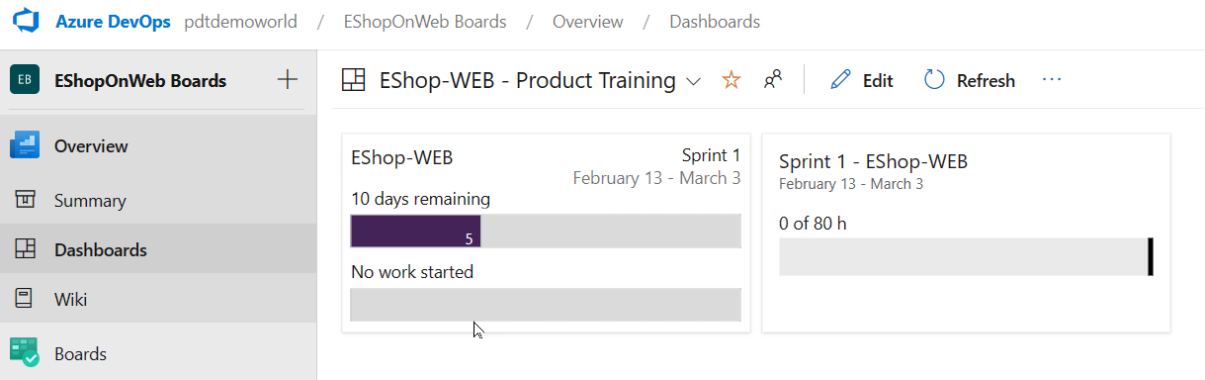
Note: The customization level will vary by widget.

8. On the **Configuration** pane, select **Close** without making any changes.

9. Back on the **Add Widget** pane, in the **Search** textbox, type **sprint** again to find existing widgets that focus on sprints. In the list of results, select **Sprint Capacity** and select **Add**.

Note: If the widget shows "Set capacity to use the sprint capacity widget", you can select the **Set capacity** link to set the capacity. Set the Activity to Development and the Capacity to 1. Select **Save** and back to the dashboard.

10. In the **Dashboard** view, at the top of the pane, select **Done Editing**.



Note: You can now review two important aspects of your current sprint on your custom dashboard.

Note: Another way of customizing dashboards is to generate charts based on work item queries, which you can share to a dashboard.

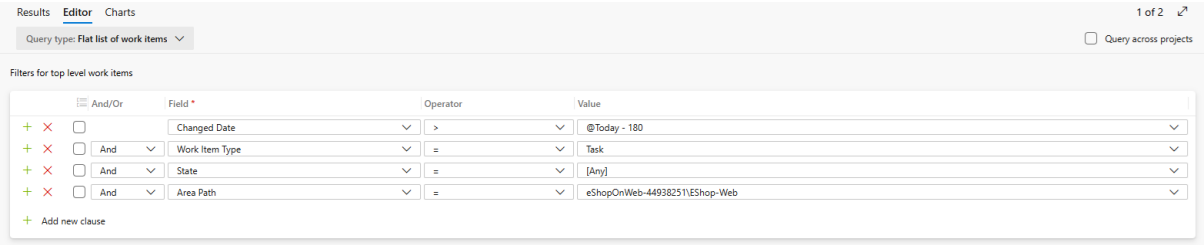
11. In the vertical navigational pane of the Azure DevOps portal, select the **Boards** icon and, in the list of the **Boards** items, select **Queries**.

12. On the **Queries** pane, select **+ New query**.

13. On the **Editor** tab of **Queries > My Queries** pane, in the **Value** dropdown list of the **Work Item Type** row, select **Task**.

14. Select **Add new clause**, and in the **Field** column, select **Area Path** and, in the corresponding **Value** dropdown list, select **eShopOnWeb\EShop-Web**.

15. Select **Save**.

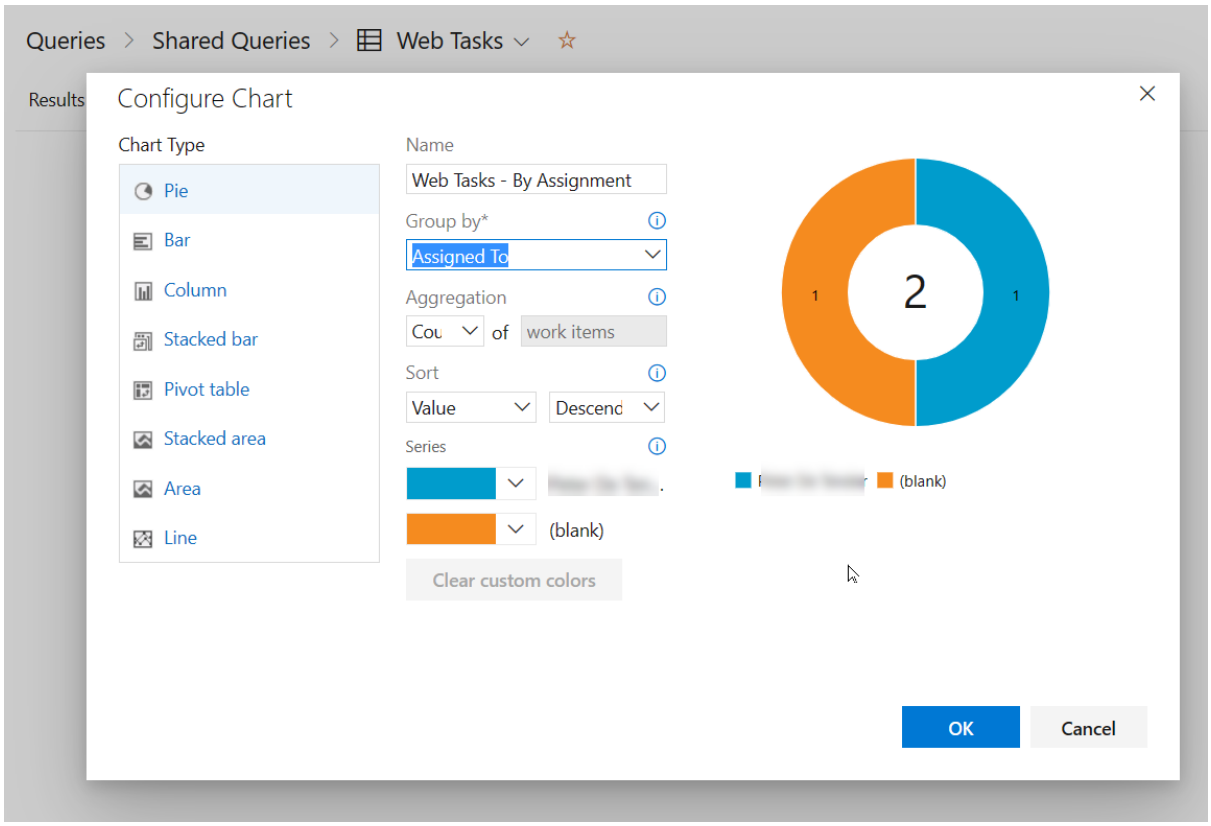


16. In the **Enter name** textbox, type **Web tasks**, in the **Folder** dropdown list, select **Shared Queries**, and select **OK**.

17. On the **Queries > Shared Queries** pane, select **Web tasks** to open the query.

18. Select the **Charts** tab and select **New chart**.

19. On the **Configure Chart** panel, in the **Name** textbox, type **Web tasks - By assignment**, in the **Group by** dropdown list, select **Assigned To**, and select **Save chart** to save the changes.



Note: You can now add this chart to a dashboard.

20. Return to the **Dashboards** section in the **Overview** Menu. From the **EShop-Web** section, select the **Product Training** dashboard you used earlier, to open it.

21. Select **Edit** from the top menu. From the **Add Widget** list, search for **Chart** , and select **Chart for Work Items**. Select **Add** to add this widget to the EShop-Web dashboard.
22. Select the **configure** (cogwheel) within the **Chart for Work Items** to open the widget settings.
23. Accept the title as is. Under **Query**, select **Shared Queries / Web Tasks**. Keep **Pie** for Chart Type. Under **Group By**, select **Assigned To**. Keep the Aggregation (Count) and Sort (Value / Ascending) defaults.
24. Confirm the configuration by selecting **Save**.
25. Notice the query results pie chart is shown on the dashboard. **Save** the changes by pressing the **Done Editing** button on top.

Clean up resources

You don't need to clean up your Azure DevOps organization or project, as they will remain available for you to use as a reference and portfolio item. Azure DevOps provides free tier usage that includes basic features for small teams.

If you want to delete the project, you can do so by following these steps:

1. In your browser navigate to the Azure DevOps portal at https:*//aex.dev.azure.com*.
2. Navigate to the **eShopOnWeb** project you created.
3. On the project settings page, go to **Overview** and select **Delete** at the bottom of the page.
4. Type the project name to confirm deletion and select **Delete**.

CAUTION: Deleting a project deletes all work items, repositories, builds, and other project artifacts. If you used an existing project for this exercise, any existing resources outside the scope of this exercise will also be deleted.

Congratulations

You have successfully completed this lab. Click **End** to mark the lab as **Complete**.

1